$$\frac{d}{dx}(f(x)) \leq 0, \forall x \in (a,b). \quad \ldots(3)$$

(4) $f(x)$ is strictly decreasing in the interval (a,b) if

$$\frac{d}{dx}(f(x)) < 0, \forall x \in (a,b). \quad \ldots(4)$$

Remark

It is very important to note the following fact. *It is* false *to say that if a differentiable function* $f(x)$ *on* I *is strictly increasing on* I, *then* $f'(x) > 0$ *for all* $x \in I$. For instance, consider $y = x^3$, $x \in (-\infty, \infty)$. It is strictly increasing on $(-\infty, \infty)$. To prove this, let $a > b$. Then we have to prove that $f(a) > f(b)$. For this purpose, we have to prove $a^3 - b^3 > 0$.

Now,

$$\begin{aligned} a^3 - b^3 &= (a-b)(a^2 + ab + b^2) \\ &= (a-b)\frac{1}{2}(2a^2 + 2ab + 2b^2) \\ &= (a-b)\frac{1}{2}\left((a+b)^2 + a^2 + b^2\right) \\ &> 0 \text{ since } a - b > 0 \text{ and other terms inside the bracket are } > 0. \end{aligned}$$

Hence it is clear that the quadratic expression is always positive (it is equal to zero only if $a = b = 0$, which contradicts the condition $a < b$). Therefore the function is $y = x^3$ is strictly increasing in $(-\infty, \infty)$. But $f'(x) = 3x^2$ which is equal to zero at $x = 0$.

Definition 7.6

A stationary point $(x_0, f(x_0))$ of a differentiable function $f(x)$ is where $f'(x_0) = 0$.

Definition 7.7

A critical point $(x_0, f(x_0))$ of a function $f(x)$ is where $f'(x_0) = 0$ or does not exist.

Note

We State that if (x,y) is a Stationary point or Critical Point of f where x from the domain of f is called Stationary number or Critical number

Every stationary point is a critical point however all critical points need not be stationary points. As an example, the function $f(x) = |x - 17|$ has a critical point at $(17, 0)$ but $(17, 0)$ is not a stationary point as the function has no derivative at $x = 17$.

Example 7.46

Prove that the function $f(x) = x^2 + 2$ is strictly increasing in the interval $(2, 7)$ and strictly decreasing in the interval $(-2, 0)$.

Solution

We have,

$$\begin{aligned} f'(x) &= 2x > 0, \forall x \in (2,7) \text{ and} \\ f'(x) &= 2x < 0, \forall x \in (-2,0) \end{aligned}$$

and hence the proof is completed. ■

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